Enhancing Genetic Ancestry Inference through Topological Data Analysis and Unsupervised Clustering

Justin Wu[1] & Tianbo Yang[2]

[1]Carnegie Mellon University (jzwu@andrew.cmu.edu), [2]University of Pennsylvania (yangtianbo11@gmail.com)

Abstract

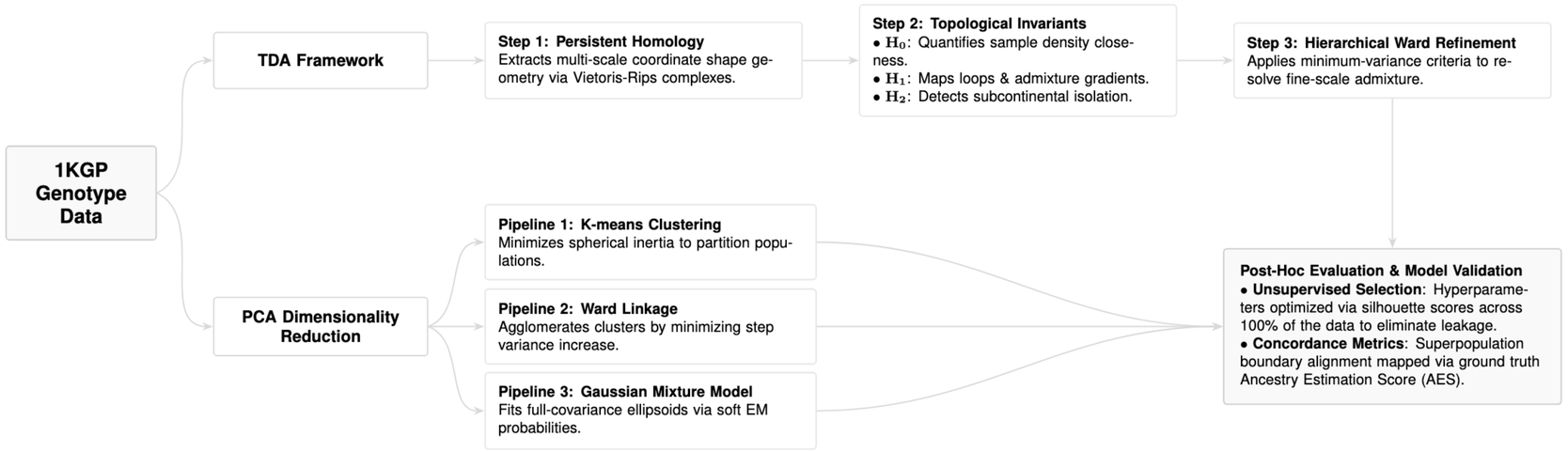
Genetic ancestry inference methods have grown increasingly important with the spread of computational medicine. Yet, current methods heavily rely on linear dimensionality reduction techniques that struggle to distinguish admixed populations. Topological Data Analysis (TDA) offers a novel approach to this issue by utilizing the nonlinear shapes of high-dimensional data that can capture features linear methods cannot.

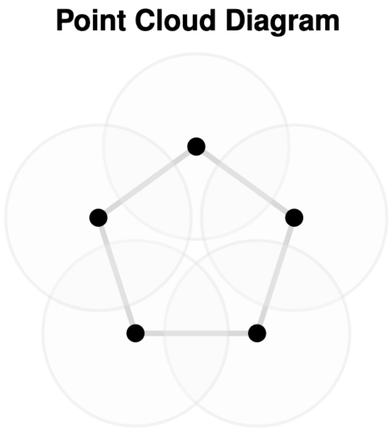
This work analyzes single nucleotide polymorphism (SNP) data from the 1000 Genomes Project (1KGP) using persistent homology. The extracted features are consistent with known demographic properties: African populations possess the highest genetic diversity, admixed American populations show simplified topological structures originating from recent admixture, and East Asian, European, and South Asian populations demonstrate high topological complexity indicative of strong regional differentiation.

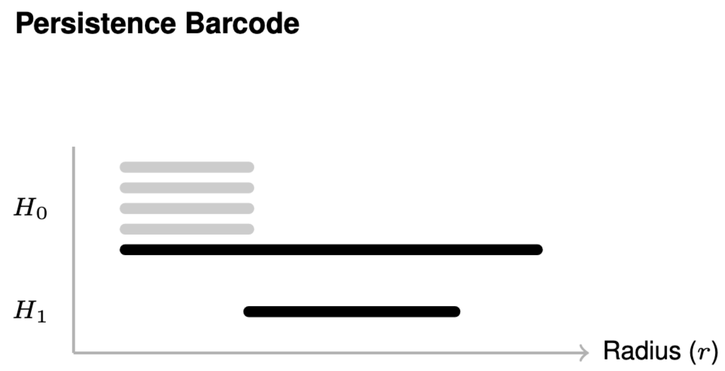
Four clustering pipelines are investigated, using silhouette scores for unsupervised hyperparameter selection to prevent data leakage. Using only topological features, TDA with Ward showcases similar accuracy (99.08%) to that of PCA with GMM, which obtained the highest accuracy (99.12%). Across all methods, the admixed American population remains the primary classification challenge due to its continuous genetic gradient.

These results indicate that persistent homology successfully interprets biologically meaningful population structures and builds upon current ancestry inference methodologies by integrating previously inaccessible nonlinear population features.

Methodology



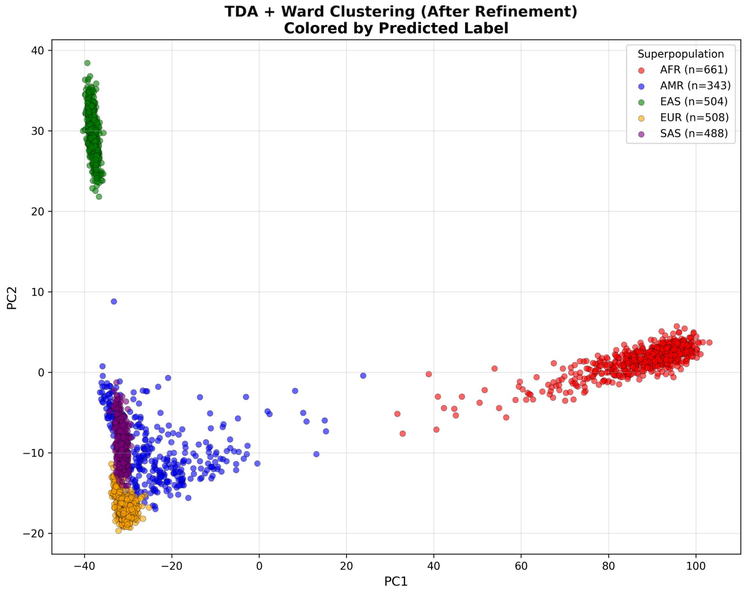




Results

Performance Comparison of Clustering Methods

Method	Silhouette	AES (%)	Time (s)
PCA + K-means	0.7312	95.93	0.98
PCA + Ward	**0.7721**	66.57	**0.79**
PCA + GMM	0.7152	**99.12**	1.00
TDA + Ward	0.6492	99.08	54.59



Conclusion

- Persistent homology features (H_0, H_1, H_2) successfully extract distinct population signatures that accurately reflect known demographic histories and migration timelines.
- African populations exhibit the highest filtration threshold (55.42), structurally confirming maximum genetic diversity as the human ancestral origin.
- Admixed American populations display a highly simplified topology (H_1 = 31, H_2 = 5), capturing a continuous genetic gradient from recent historical mixing rather than discrete clusters.
- Modeling continuous ancestry gradients remains a critical frontier, as admixed groups present the primary classification boundary challenge across all architectural models.
- Unsupervised topological features capture non-linear global connectivity structures that linear dimensionality reduction techniques miss, providing a scalable framework for fine-scale ancestry estimation.
- Future iterations will explore ensemble architectures and alternative distance metrics, such as Mahalanobis distance, to account for anisotropic feature distributions.

References

- The 1000 Genomes Project Consortium. (2015). A global reference for human genetic variation. Nature, 526(7571), 68–74. https://doi.org/10.1038/nature15393
- Bauer, U. (2021). Ripser: efficient computation of Vietoris–Rips persistence barcodes. Journal of Applied and Computational Topology, 5(3), 391–423. https://doi.org/10.1007/s41468-021-00071-5
- Amézquita, E. J., Quigley, M. Y., Ophelders, T., Munch, E., & Chitwood, D. H. (2020). The shape of things to come: Topological data analysis and biology, from molecules to organisms. Developmental Dynamics, 249(7), 816–833. https://doi.org/10.1002/dvdy.175
- Price, A. L., Patterson, N. J., Plenge, R. M., Weinblatt, M. E., Shadick, N. A., & Reich, D. (2006). Principal components analysis corrects for stratification in genome-wide association studies. Nature Genetics, 38(8), 904–909. https://doi.org/10.1038/ng1847
- Dempster, A. P., Laird, N. M., & Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. Journal of the Royal Statistical Society: Series B (Methodological), 39(1), 1–22. https://doi.org/10.1111/j.2517-6161.1977.tb01600.x

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